João Pedro Peters Barbosa

joao.peters@ieee.org | +55 32 99952-6401 | joaoppeters.github.io
Av. Francisco Pereira Lopes, 1950, Ap 801/A, Sao Carlos, SP, Brazil

EDUCATION

USP - EESC
PhD in Electrical Engineering
Expected 2027 | São Carlos, SP
Emphasis on Energy Systems

UFJF
MSc in Electrical Engineering
May 2023 | Juiz de Fora, MG
GPA: 3.75 / 4.0
Emphasis on Energy Systems

UFJF
BSc in Electrical Engineering
April 2021 | Juiz de Fora, MG
IRA: 78.47 / 100
Power Systems Electrical Engineer

TEMPLE UNIVERSITY
Academic Exchange
Dec 2019 | Philadelphia, PA
GPA: 3.74 / 4.0

SKILLS

PROGRAMMING
Python • C++ • C • Matlab
Git • GitHub • LaTeX

POWER SYSTEMS TOOLS
ANAREDE • ANATEM • PSS/E
PSCAD • OpenModelica
OpenDSS • PSIM

LANGUAGES
Portuguese (Native)
English (Fluent - CAE/FCE/TOEFL)
French, German, Spanish (Basic)

LINKS

Github:// **joaoppeters**
LinkedIn:// **joaoppeters**

CERTIFICATES

Scientific Computing (Python)
Data Analysis (Python)
Intro to Git/GitHub
OpenDSS Simulation

EXPERIENCE

SIEMENS | Power Systems Consultant Engineer
April 2026 – Present | Brazil

USP - EESC | Graduate Research Assistant
June 2023 – Present | São Carlos, SP

- Researching electrical power systems stability and control.

UFJF | Graduate Research Assistant
May 2021 – May 2023 | Juiz de Fora, MG

- Research on Smooth Power Flow (SPF) formulation for voltage stability.

UFJF & PETROBRAS | Undergraduate Research Assistant
March 2020 – April 2021 | Juiz de Fora, MG

- Research on Distributed Energy Resources (DERs) to provide grid services.

UFJF | Undergraduate Research Assistant
Aug 2016 – July 2019 | Juiz de Fora, MG

- **Control Systems:** Applied PID techniques to buck-boost converters.
- **MMC Modules:** Research on sorting methods for MMC modules.

KEY PROJECTS & PUBLICATIONS

PYANA | Open-source Tool
Python-based library developed to support researchers in steady-state analysis of power systems by processing data from ANAREDE files.

RECENT PUBLICATIONS

- **IEEE PES IM (2026):** Curtailment Mitigation in Wind Power Integration.
- **JMPCE Journal (2025):** Definition of contingency criticality based on violation probability of voltage security margin.
- **IEEE Access Journal (2025):** Voltage stability assessment through modal analysis using a smooth formulation based on sigmoid functions.

HONORS/AWARDS

2022	Best Article (Master's)	SBSE Conference
2017	Best Article (Undergrad)	COBEP Conference
2016	Best Article (Undergrad)	SEMIC Conference

MEMBERSHIPS

2024 – Present	Member, CIGRé
2022 – Present	Member, Sociedade Brasileira de Automática (SBA)
2020 – Present	Member, IEEE Young Professionals Society (YP)
2021 – Present	Registered Engineer, CREA-MG
2018 – Present	Member, IEEE Power and Energy Society (PES)
2016 – Present	Member, IEEE